

DATA ANALYSIS SUMMARY REPORT

1. INTRODUCTION

This report analyzes two datasets: the Titanic dataset and the Housing dataset. The aim is to explore patterns, perform statistical analysis, and validate findings using hypothesis testing.

2. TITANIC DATASET SUMMARY

Descriptive Statistics:

- Average passenger age is approximately 29 years.
- Survival rate is about 38%, while 62% of passengers did not survive.

Key Insights:

- Gender: Females had significantly higher survival rates than males.
- Class: First-class passengers had the highest survival rate, while third-class had the lowest.
- Fare: Higher ticket fare is associated with higher survival probability.

Hypothesis Testing:

- Gender vs Survival: $p < 0.05$, indicating a significant relationship.
- Class vs Survival: $p < 0.05$, showing that passenger class significantly affects survival.

Correlation:

- Strong correlation between Sex and Survival.
- Moderate correlation between Fare and Survival.
- Weak correlation between Age and Survival.

Conclusion:

Survival on the Titanic was influenced mainly by gender, socio-economic class, and wealth.

3. HOUSING DATASET SUMMARY

Descriptive Statistics:

- Prices range from approximately 7,000,000 to 13,300,000.
- Most houses have 3 to 4 bedrooms and 2 bathrooms.

Key Insights:

- Area: Larger houses tend to have higher prices.
- Location: Houses in preferred areas have higher prices.
- Features: Air conditioning and furnishing increase property value.

Hypothesis Testing:

- Area vs Price: Strong positive correlation, $p < 0.05$.
- Preferred Area vs Price: $p < 0.05$, indicating significant price differences.
- Furnishing Status: ANOVA test shows $p < 0.05$, indicating differences between groups.

Correlation:

- Strong correlation between Area and Price.
- Moderate correlations with Bathrooms and Parking.
- Weak correlation with number of Bedrooms.

Conclusion:

House prices are largely influenced by size, location, and available amenities.

4. CORRELATION VS CAUSATION

Correlation indicates relationships between variables, while causation explains why those relationships exist. For example, higher fares are correlated with higher survival in the Titanic dataset, but the true cause is access to better resources.

5. OVERALL CONCLUSION

- Titanic: Survival depends on gender, class, and wealth.
- Housing: Price depends on area, location, and amenities.
- Statistical testing confirmed the significance of these relationships.

6. RECOMMENDATIONS

- Use correlation to identify relationships.
- Apply hypothesis testing to validate findings.
- Avoid confusing correlation with causation.